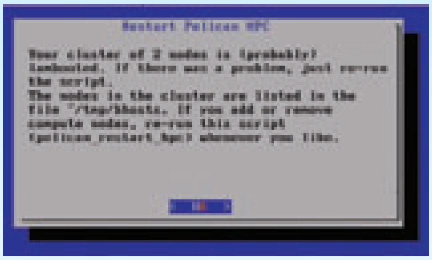
Supercomputer drivers?
A new supercomputer called NeuFlow is capable of driving a car efficiently through the use of a human based visual system

Long road ahead
India owns a measly 1% in the share of the world's top 500 supercomputers

Did you know?
A single modern desktop PC is now more powerful than a ten year old supercomputer

Does it work?!?

Now to test your cluster using some regression calculations. Start terminal and type in "octave" and press enter. Now type "kernel_example" and press enter (drumroll please) to see some really cool econometrics curves generated by your cluster. You may also want to try other examples such as "pea_example", "bfgsmin_example", "mle_example", "gmm_example", "mc_example1" and "mc_example2".



Hit "No" till all compute nodes are accounted for.

Screen after completion of netboot

You also get a time taken to generate as one of the outputs in the terminal. This can serve as your first benchmark for how much faster your new cluster is performing.

How much faster is your new Super rig?

Alright, so now you are the new cool kid on the block with loads of super computing power at your disposal. But what speeds are you really touching? Well, a simple way of knowing how much faster you are going will show up right in the terminal where you ran the pelican_setup command. Once you are done connecting all the compute nodes and hit "Yes", just wait in terminal to get an overview of what speeds you are achieving and how much each compute node is contributing.

Bear in mind that the speed shown here to you is not optimised for your particular cluster since all the compute nodes (irrespective of the number of cores you have) will only receive one MPI rank, hence limiting the work of each compute node to one core. This can be optimised to properly balance the load of an application, but the optimisation is left to the user. You can find help on the PelicanHPC forum with regard to this if you are really serious about getting maximum compute power out of your cluster.

Problem with measuring cluster performances is that there is no general purpose tool available for testing all components. Currently, the LINPACK tool is used for cluster benchmarking, but it does not measure all cluster features. Head over to `http://bit.ly/9QyPU6` to get your fix of all the cluster benchmarks you need to know about.

Three finger salute

Though clusters are not popular in the home environment these days, the light at the end of the tunnel seems very bright. With ongoing research on sharing GPU load within the cluster and software developers using the OpenMPI application programming interface in their software development process, cluster variants of our favourite software and games seem possible in the not so distant future. Look to cluster computing as an opportunity to innovate and come up with your own implementation of load balancing over the network.

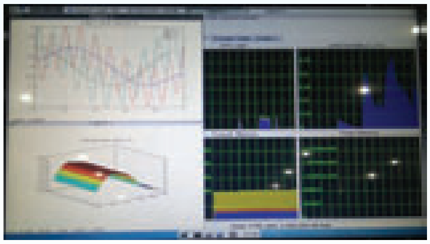
With all the information required freely available on the internet (Open Source), students can use this as a research subject and base their projects on this new world of parallel computing. Quoting the father of our nation, "Be the change you want to see in the World" and look to the horizon to make the world of computing a faster, better place.

Food for thought

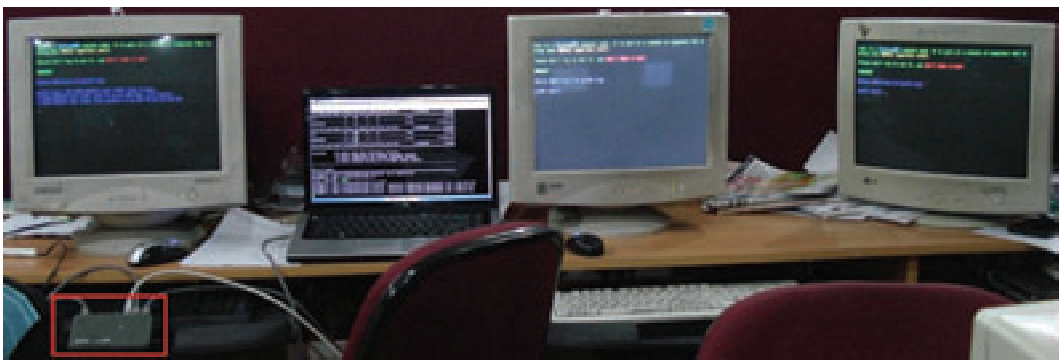
Gaming on clusters though not possible right now, is a very mouth watering prospect wherein old PC's with legacy onboard graphics would theoretically be able to use the graphics rendering from another system on the cluster. This would mean LAN parties for everyone, provided some of the systems on the cluster have a powerful enough GPU. This would also mean that games need to be designed to use what is known as MPI such that GPU resources can be offloaded and shared with other systems in the cluster.

Hook, line and sinker!

When James Cameron was filming. The Titanic (remember Kate Winslet?), his special effects crew could not afford a supercomputer to do the rendering. Hence, a massive cluster was setup and all the water related critical rendering was performed on that cluster because as the saying goes, many hands make for light work.



ksysguard monitoring cpu load on the cluster



4 computers on a cluster with the switch highlighted